

PRANAV P

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SUMMARY

AI/GenAI Engineer building production-ready agentic AI systems across LLM orchestration, RAG, real-time voice and speech-to-speech interaction, tool use, OCR, evaluation, and cloud deployment. Experienced in designing shared agentic backends serving voice and text interfaces, with specialized agents for retrieval, SQL, OCR, and data tasks. Strong ML foundation across time-series, satellite imagery, and explainable ML, with hands-on AWS deployment.

EXPERIENCE

FarmwiseAI Pvt. Ltd. — Data Scientist

Jul 2026 – Present

Perungudi, Chennai

Associate Data Scientist

Apr 2025 – Jun 2026

- Lead a team of 3 data scientists while continuing as an individual contributor, coordinating technical execution and delivery across AI/ML initiatives.
- Built a production-ready agentic AI platform with a shared LangChain reasoning layer powering both text and real-time voice interfaces, with specialized agents and sub-agents for knowledge retrieval, SQL, OCR, and data tasks.
- Engineered real-time voice and speech-to-speech interaction with interruption handling and asynchronous tool execution, exposing the same underlying agentic capabilities through voice and text.
- Evaluated vector and structure-aware retrieval approaches; adopted vectorless retrieval for structured documents after embedding similarity did not reliably correlate with answer relevance, accepting a latency trade-off for improved answer quality.
- Built specialized agents for knowledge retrieval, SQL analysis, and OCR, combining frontier-model OCR with self-hosted LightOnOCR-2 1B on AWS SageMaker for sensitive documents.
- Implemented model routing with LiteLLM across OpenAI, Vertex AI/Gemini, AWS Bedrock, and local models, selecting models based on task requirements, accuracy, latency, cost, and data sensitivity.
- Built agent-driven UI interactions capable of navigating application screens and surfacing cards, dashboards, calculated charts, and bar graphs based on conversational intent.
- Built a golden-dataset benchmarking workflow to compare models across accuracy, latency, and cost, enabling task-level model selection and routing.
- Independently identified an HR technical-screening bottleneck and built a Chromium browser-based AI screening workflow using JD/resume context, real-time interviewer/candidate audio separation, role-specific technical question generation, and response evaluation before the technical interview.
- Deployed AI services on AWS using ECS/Fargate and SageMaker, including local model hosting for sensitive workloads.

Infosys — Data Visualization and Analyst Engineer Intern

Feb 2025 – Apr 2025

- Built Power BI and Python-based dashboards, performing data preprocessing, exploratory analysis, and trend analysis to support business reporting.

Descpro — ML Developer Intern

Dec 2023 – Apr 2024

- Developed ML solutions for an NBFC loan approval and monitoring workflow, including eligibility modeling, LSTM-based repayment-delay prediction, and loan inquiry forecasting.

- Developed predictive clinical risk models using engineered health indicators and LSTM-based alerting, with SHAP/LIME for model interpretation.

SELECTED PROJECTS

Domain-Specific Loan Officer Agent

Gemma-2B | QLoRA | Unsloth | llama.cpp

- Fine-tuned a small language model on a synthetic instruction dataset using QLoRA and Unsloth, converted the model to GGUF, and deployed it with llama.cpp for low-latency CPU inference.

TECHNICAL SKILLS

GenAI & Agentic Systems: Agentic AI, RAG, Structure-Aware/Vectorless RAG, LangChain, LangGraph, Google ADK, MCP, Tool Calling, Pydantic AI, Instructor, LiteLLM, Langfuse, Model Routing

LLM & AI Infrastructure: OpenAI, Gemini/Vertex AI, AWS Bedrock, Hugging Face, vLLM, Ollama, Unsloth, QLoRA, SageMaker, ECS/Fargate, FastAPI, WebSockets

Machine Learning: Random Forest, LSTM, Time-Series Analysis, Dynamic Time Warping, Feature Engineering, SHAP, LIME, Data Interpolation, Leakage Prevention

Data & Engineering: Python, SQL, Pandas, NumPy, PySpark, Qdrant, FAISS, Streamlit, Power BI, Git, Satellite Imagery Processing

EDUCATION

VIT Chennai

B.Tech — Computer Science & Engineering, AI & ML Specialization
CGPA: 7.98

2021 – 2025